

EX No : 2

EXPLORATORY DATA ANALYSIS - DATA IMPORT AND EXPORT

Aim

To analyze the Car Prediction dataset using Exploratory Data Analysis (EDA), identify important factors affecting used-car selling price, visualize fuel type and seller type performance, build a machine learning model to predict selling price, evaluate the model, and export the processed data to Excel.

Dataset Overview

Dataset file: car_prediction_data.csv

Dataset shape: 301 rows $\times$ 9 columns

Columns: Car_Name, Year, Selling_Price, Present_Price, Kms_Driven, Fuel_Type, Seller_Type, Transmission, Owner

Target variable: Selling_Price

Average Selling Price: 4.66

Minimum Selling Price: 0.10

Maximum Selling Price: 35.00

Missing values: 0

Procedure

1. Load the Car Prediction dataset using Pandas.
2. Display the first few records, shape, information, and missing values.
3. Check and handle missing values.
4. Create a Car_Age feature from the Year column.
5. Perform EDA using descriptive statistics and group-wise analysis.
6. Analyze average selling price by Fuel Type, Seller Type, and Transmission.
7. Examine relationships between Present Price, Kms Driven, and Selling Price.
8. Prepare numerical and categorical features for machine learning.
9. Split the dataset into training and testing data.
10. Train a Random Forest Regression model to predict Selling Price.
11. Predict selling price for the test dataset.
12. Evaluate the model using MAE, RMSE, and R^2 score.
13. Export the processed dataset to Excel.

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
```

```

from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# 1. Load dataset
df = pd.read_csv("car_prediction_data.csv")

# 2. Display data and information
print(df.head())
print(df.shape)
print(df.info())
print(df.isnull().sum())

# 3. Handle missing values
df = df.copy()

# 4. Create Car Age feature
df["Car_Age"] = 2026 - df["Year"]

# 5. EDA
print(df.describe())
print(df.groupby("Fuel_Type")["Selling_Price"].mean().sort_values(ascending=False))
print(df.groupby("Seller_Type")["Selling_Price"].mean().sort_values(ascending=False))
print(df.groupby("Transmission")["Selling_Price"].mean().sort_values(ascending=False))

# 6. Visualization
df.groupby("Fuel_Type")["Selling_Price"].mean().sort_values(ascending=False).plot(kind="bar")
plt.title("Average Selling Price by Fuel Type")
plt.xlabel("Fuel Type")
plt.ylabel("Average Selling Price")
plt.show()

df.groupby("Seller_Type")["Selling_Price"].mean().sort_values(ascending=False).plot(kind="bar")
plt.title("Average Selling Price by Seller Type")
plt.xlabel("Seller Type")
plt.ylabel("Average Selling Price")
plt.show()

# 7. Features and target
X = df.drop(columns=["Selling_Price"])
y = df["Selling_Price"]

num_cols = X.select_dtypes(include=np.number).columns.tolist()
cat_cols = X.select_dtypes(exclude=np.number).columns.tolist()

# 8. Preprocessing
preprocessor = ColumnTransformer([
    ("num", SimpleImputer(strategy="median"), num_cols),
    ("cat", Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore"))
    ]), cat_cols)
])

# 9. Random Forest Regression
model = Pipeline([
    ("preprocessor", preprocessor),
    ("regressor", RandomForestRegressor(
        n_estimators=200, random_state=42
    ))
])

# 10. Train-test split
X_train, X_test, y_train, y_test = train_test_split(

```

```

X, y, test_size=0.2, random_state=42
)

# 11. Train
model.fit(X_train, y_train)

# 12. Predict
y_pred = model.predict(X_test)

# 13. Evaluation
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)

# 14. Export
df.to_excel("Car_Prediction_Processed.xlsx", index=False)
print("Analysis completed!")

```

Output

First five records:

Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
swift	2014	4.6	6.87	42450	Diesel	Dealer	Manual	0

Dataset Shape: (301, 9)

Missing Values: No missing values were found in the supplied dataset.

Average Selling Price: 4.66

Minimum Selling Price: 0.10

Maximum Selling Price: 35.00

Correlation between Present Price and Selling Price: 0.8790

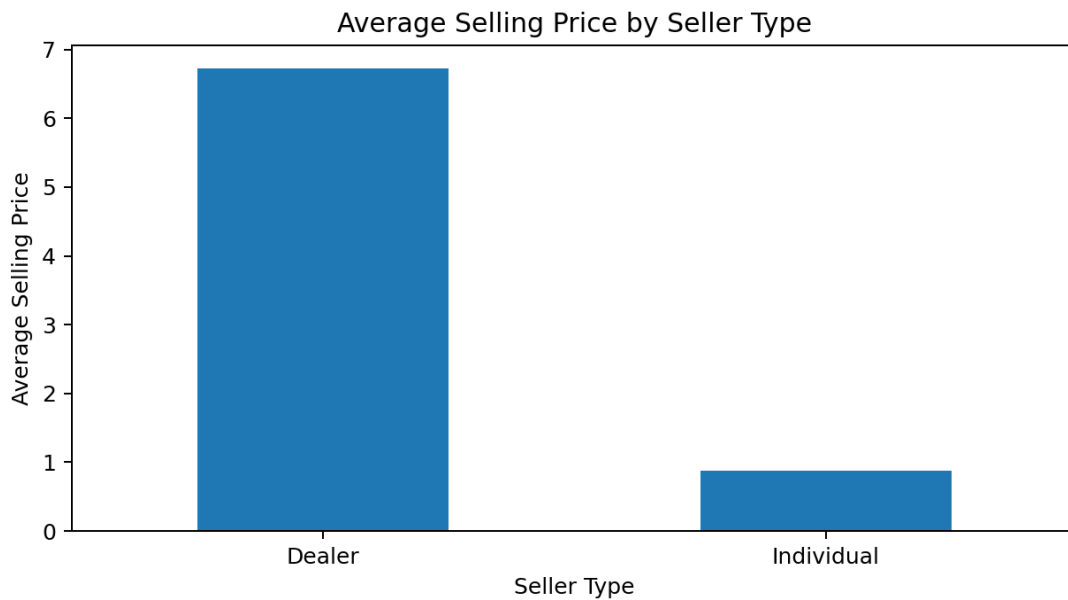
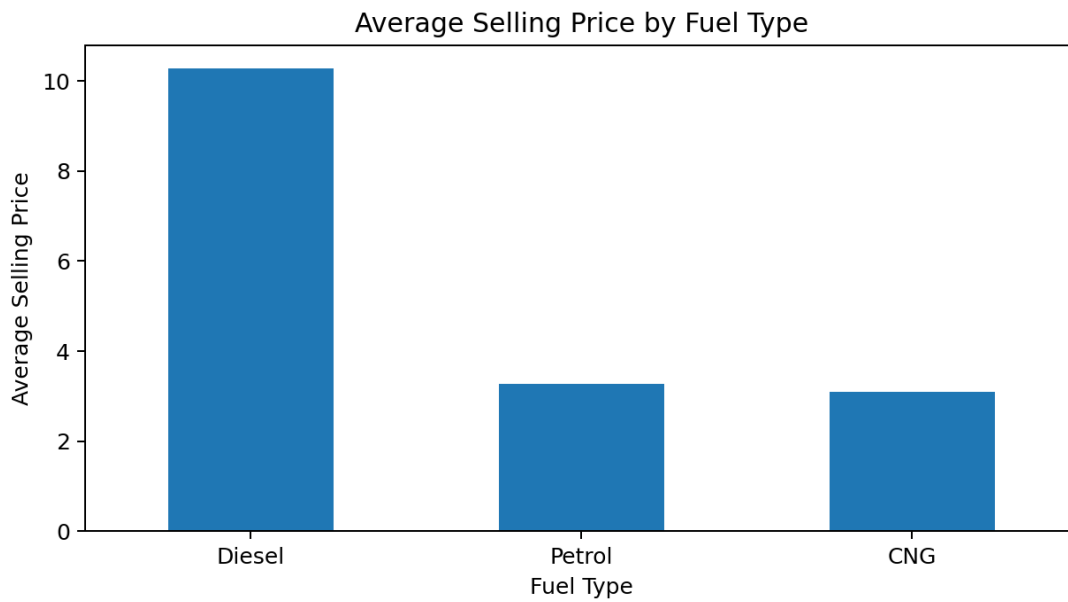
Correlation between Kms Driven and Selling Price: 0.0292

EDA Results

Fuel type with the highest average selling price: Diesel

Seller type with the highest average selling price: Dealer

Transmission type with the highest average selling price: Automatic



Model Evaluation

Training records: 240

Testing records: 61

MAE: 0.5901

RMSE: 0.8872

R² Score: 0.9658

Result

The Car Prediction dataset was successfully imported, explored, processed, and modeled. The dataset contains 301 records and 9 original fields. EDA was performed on fuel type, seller type, transmission, present price, and kilometres driven. A Car_Age feature was derived from the Year column. A Random Forest Regression model

was trained to predict Selling_Price and achieved an R^2 score of 0.9658 on the test data. The processed dataset was exported successfully to Car_Prediction_Processed.xlsx.